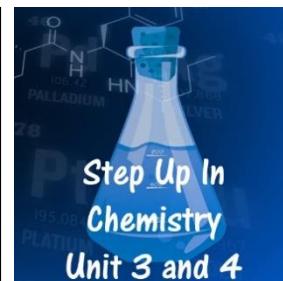


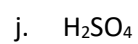
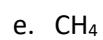
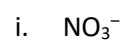
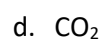
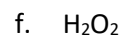
3.1 Redox Reactions and Oxidation Numbers

Problems Worksheet

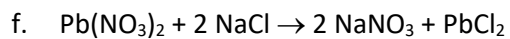
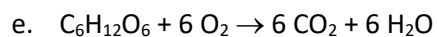
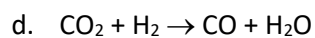
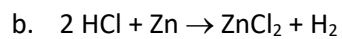
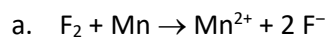


1. Explain how the modern definition of oxidation differs from traditional definitions.
2. Use the example of sodium metal reacting with chlorine gas to explain the concept of oxidation and reduction.

3. Determine the oxidation number of each of the elements in the following:



4. Use oxidation numbers to determine which species are oxidised and reduced in the following equations:



5. Bromate is an ion with the formula BrO_3^- where bromine has an oxidation state of +5. Determine the formula of the ion, and the oxidation state of bromine in the following:
- Bromide
 - Bromite
 - Perbromate
 - Hypobromite